Snehit Chunarkar

ML Researcher | Noise Robust Speech Emotion

(+886) 0965742974 Email LinkedIn Website GitHub Hsinchu, Taiwan

Summary

ML Researcher focused on noise-robust speech emotion recognition and multimodal audio-text learning | PhD at NTHU's BIIC Lab under Prof. Chi-Chun Lee (Jeremy).

Research focus: speech/audio representation learning, emotion understanding, and cross-modal fusion (audio–text), with an emphasis on robust modeling and interpretability.

Experience with top-tier models: WavLM, Wav2Vec2, CLAP, Whisper, RoBERTa. Globally #2 in ICASSP 2026 SP Grand Challenge (XACLE) with 92% better prediction; led a noise-robust speech emotion recognition system with 32% improvement.

Education

PhD, Electrical Engineering (2027 expected)
National Tsing Hua University (NTHU), Hsinchu, Taiwan

M.Tech., Instrumentation and Signal Processing
Indian Institute of Technology (IIT), Roorkee, India

B.E., Instrumentation Engineering
Government College of Engineering, Chandrapur, India

Research Advisor

Prof. Chi-Chun Lee (Jeremy)
Professor & Associate Chair, NTHU; Research Fellow, Academia Sinica; Director, NVIDIA–NTHU Joint Innovation Center

Publications

Reasoning Driven Captions to Assist Noise Robust Speech Emotion Recognition
Snehit B. Chunarkar, Chi-Chun Lee - IEEE ICASSP 2026 (Barcelona, Spain)

Cross-Modal Semantic Alignment Via Ensemble Audio-Text Features for XACLE Challenge
Snehit B. Chunarkar, Krishnagiri Hamza, Chi-Chun Lee - IEEE ICASSP Workshop 2026 (Barcelona, Spain)

Exploring Pretrained Audio-Text Encoders for Audio Moment Retrieval: DCASE 2026 Task 6
Snehit B. Chunarkar, Krishnagiri Hamza, Chi-Chun Lee - DCASE2026 (Boston, USA)

STELIN-US: A Spatio-Temporally Linked Neighborhood Urban Sound Database
S. Chunarkar, B. Su, C. Lee - DCASE2023 (Tampere, Finland)

Structured Event-to-Text Generation for Zero-Shot Audio Description: A Prompt Study Across LLM Sizes
Snehit B. Chunarkar, Chi-Chun Lee - Preprint: ResearchGate

Mixed Language Separation Using Deep Neural Network
S. Chunarkar, S. Chiluveru, M. Tripathy - ICEECCOT 2021, (Mysuru, India)

Efficient Hardware Implementation of Nonlinear Activation Function for Inference Model
S. Chunarkar, S. Chiluveru - ISOCC 2024, (Sapporo, Japan)

Projects

Reasoning-driven-SER: (Speech Emotion Recognition in Noisy Environment)

ICASSP 2026 | Most Competitive year with only 40% acceptance rate

Integrating Reasoning-enriched caption and Audio using Cross-Attention to enhance Speech Emotion Recognition (SER) in Noisy Environment.

Improved SER by 32% over the conventional Audio-Only approach at SNR: -10dB.

EnsembleSVR-XACLE: (Audio-Text Alignment Prediction)

ICASSP 2026 SP Grand Challenge: XACLE | Rank #2 (Globally)

Engineered end-to-end pipeline to predict audio-text alignment scores in multimodal datasets

Ranked #2 on official leaderboard among 40+ submissions

AMR Encoder Exploration: (Audio Moment Retrieval)

DCASE 2026 Challenge: Task 6 | Top 12 (Globally)

Explored Audio-Text Encoders for Audio Moment Retrieval (AMR)

116.66% improvement in Recall@0.7 on Evaluation data compared to Baseline - leaderboard.

STeLiN-US Database (Dataset synthesis for real-world SED)

DCASE Publication

Synthesized a dataset to mimic real-world acoustic environments (overlapping events, spatial distance scaling, temporal correlation) for training sound event detection systems.

SED2Text: (Sound Events in Natural Language)

[Preprint] ReseachGate

A training-free, 0.59 billion-parameter cascade for temporally grounded audio description.

My proposed approach can replace large audio-language models for this task.

Pawpularity Contest: (Cuteness Score Prediction for Pet's Image)

NTHU Machine Learning Course Project (11230EE 655000) | Rank #2 in class

Built ensemble regression model to predict pet adoption likelihood from image and metadata.

Ranked #2 among 16 course teams; with state-of-the-art performance on Pawpularity dataset.

Skills

Programming Languages: Python, Markdown, Bash

Machine Learning: PyTorch, NumPy, Pandas, Matplotlib, scikit-learn, Hugging Face Transformers

Development Environments: VS Code, Spyder, Jupyter Notebook, MATLAB, MobaXterm

Cloud & HPC: Google Colab, TWCC (Taiwan Computing Cloud), IIC (Innovative Island Cloud)

Additional Information

Languages: English (fluent), Hindi (native), Marathi (native)

Active Engagement: Open-source GitHub, Zenodo contributions